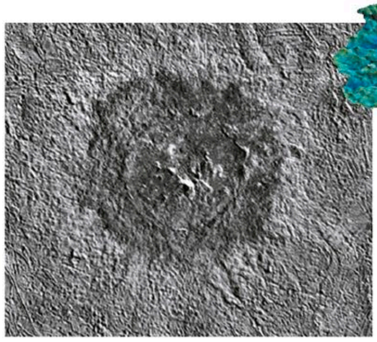
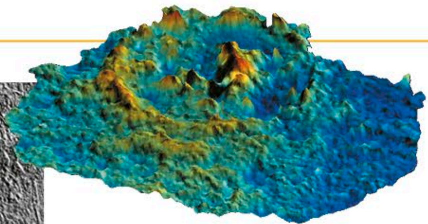
have observed the moons first, but it was Galileo who published his findings and brought the moons to the attention of the scientific and wider community.

Jupiter's fourth-largest satellite is a fascinating world. It is a little smaller than Earth's Moon but much brighter, since its icy surface reflects five times as much light. A liquid ocean may lie below Europa's water-ice crust, which is just tens of miles thick. This watery layer, which is estimated to be 50–105 miles (80–170 km) deep and to contain more liquid than Earth's oceans combined, could be a haven for life. Below lies a rocky mantle surrounding a metallic core.



OVERHEAD VIEW



TERRAIN MODEL

PWYLL CRATER
This three-dimensional model of the 16-mile (26-km) wide Pwyll Crater (above) was made by combining images (see example, left) taken from different angles and then applying color. Unusually, the crater floor (blue) is the same height as the moon's surface, and the central peak (red) is much higher than the crater's rim.

The surface appears to be geologically young and consists of smooth ice plains, disrupted terrain, and regions criss-crossed by dark linear structures that can be thousands of miles long.

The mottled appearance of the disrupted terrain comes from crust that has broken up and floated into new positions. Round or oblong, city-sized dark spots freckle the surface. Known as lenticulae, these form as large globules of warm and slushy ice push up from underneath and briefly melt the surface ice. Exactly how the dark lines formed is unclear, but volcanically heated water and ice and other kinds of tectonic activity were involved. Tidal forces fractured the crust, and liquid or icy water erupted through the crack to freeze almost instantly on the surface.

In Greek myth, Europa was the girl who was seduced by Zeus in the form of a white bull and carried off to Crete.

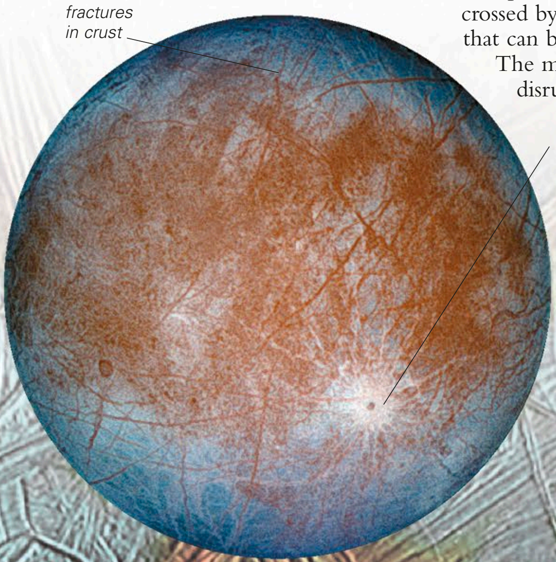
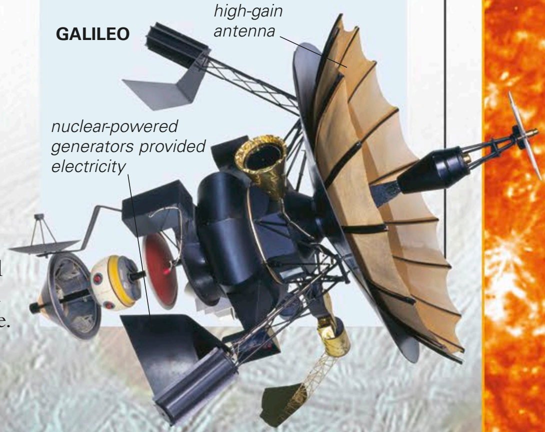


IMAGE OF THE FAR SIDE
This is how the far side of Europa would appear to the human eye. Bright plains in the polar areas (top and bottom) sandwich a darker, disrupted region of the crust.

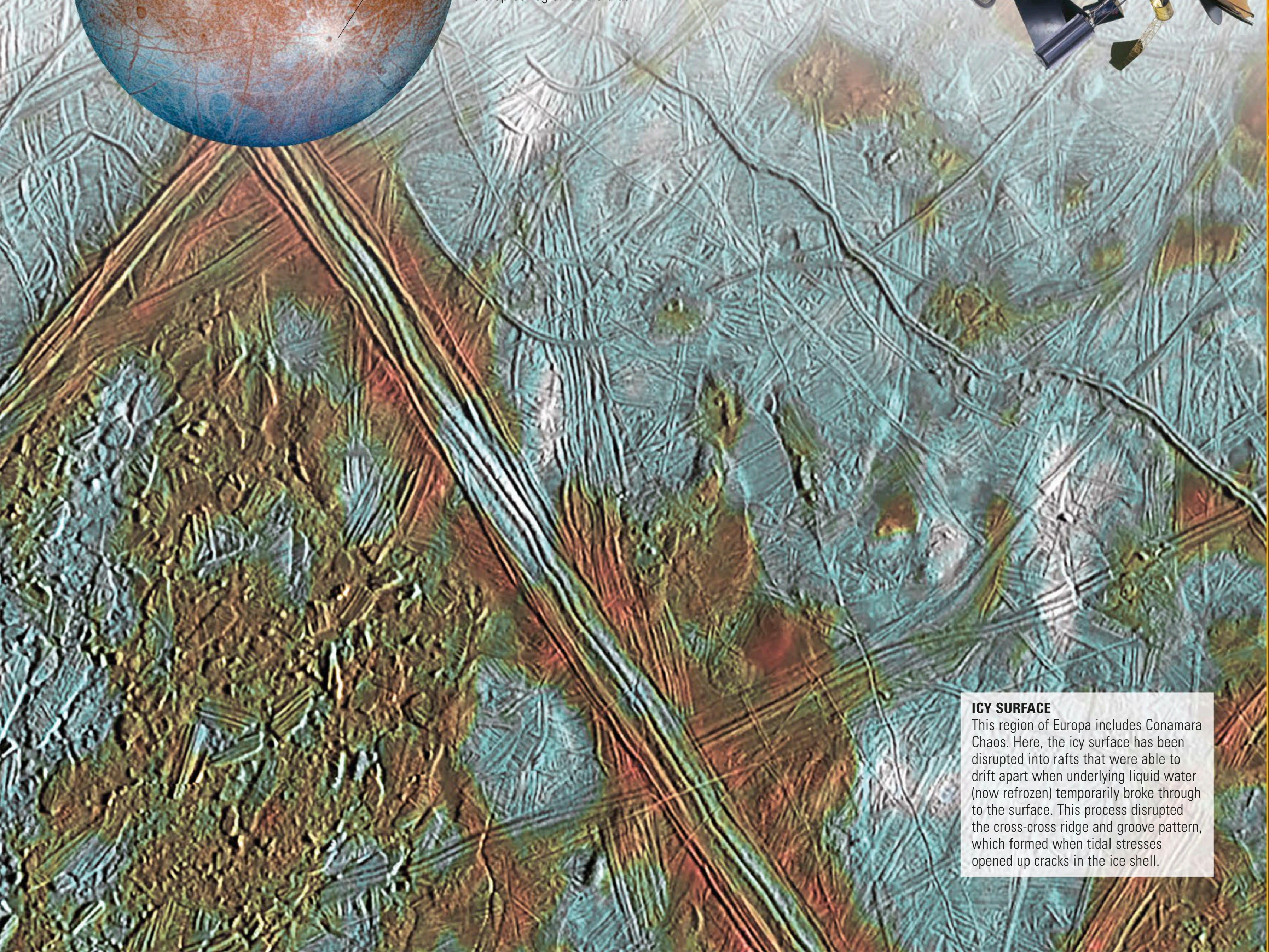
EXPLORING SPACE

KEEP EUROPA CLEAN

After a 6-year journey from Earth, the Galileo space probe spent 8 years studying the Jovian system and made 11 close flybys of Europa. The decision was made to destroy the probe because NASA wanted to avoid an impact with Europa and the potential contamination of its subsurface ocean, which could possibly harbor life. With little fuel left, Galileo was put on a collision course with Jupiter. The probe disintegrated in the planet's atmosphere on September 21, 2003.



GALILEO



ICY SURFACE
This region of Europa includes Conamara Chaos. Here, the icy surface has been disrupted into rafts that were able to drift apart when underlying liquid water (now refrozen) temporarily broke through to the surface. This process disrupted the cross-cross ridge and groove pattern, which formed when tidal stresses opened up cracks in the ice shell.